

25W1 2: Benchmarking Halpern and GRAAL beyond Theoretical Convergence Bounds on Lower Bound, Forsaken

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1 Introduction

This report presents the stability of the *inexact Halpern* and *GRAAL* methods under stressed parameter ranges where the convergence condition is not fully satisfied. Both algorithms depend critically on the relationship between the parameter ρ and the Lipschitz constant L of the underlying operator.

For Halpern, the theoretical requirement is

$$0 < \rho < \eta < \frac{1}{L},$$

while for GRAAL the stricter condition

$$\rho < \frac{1}{2L}$$

applies.

Motivated by this difference, we define three cases in order of increasing difficulty:

$$\text{Case I : } \frac{1}{L} < \rho, \quad \text{Case II : } \frac{1}{2L} < \rho \leq \frac{1}{L}, \quad \text{Case III : } \rho \leq \frac{1}{2L}.$$

We hypothesize the following behavior: in Case I both methods become unstable, typically diverging or entering limit cycles; in Case II only Halpern remains stable, while GRAAL fails to converge; and in Case III Halpern and GRAAL converge reliably.

To test this hypothesis, we tune the problem parameters in three benchmark variational inequalities **Lower Bound**, and **Forsaken**. so that ρ depend of the three cases.

Across these settings we run Halpern algorithms with stepsizes $\eta = \frac{0.99}{L}$, while GRAAL remain $\eta = \frac{2-\phi}{L}$, and evaluate convergence in terms of operator norm decay, trajectory behavior, and long run stability beyond the theoretical boundaries.

2 Experiments

2.1 Lower Bound

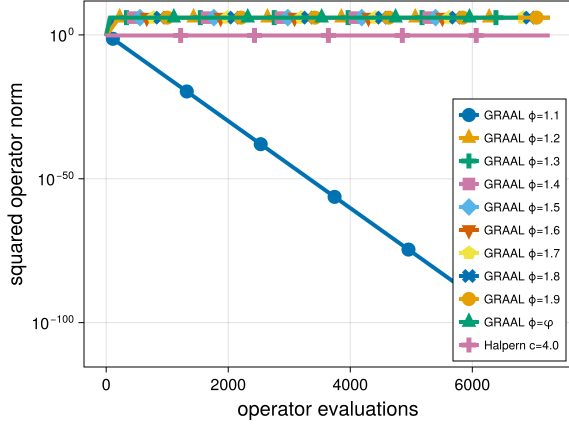
We consider the lower-bound instance with

$$\rho = -\frac{2b}{a^2 + b^2}, \quad L = \sqrt{a^2 + b^2},$$

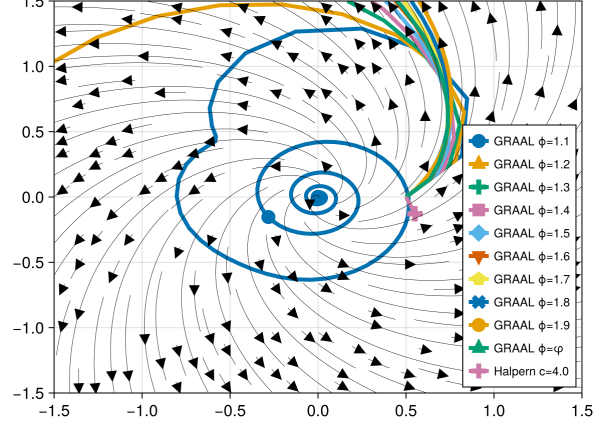
by tuning the parameter a^2 with fixed $b = -1$ in order to adjust the relation between ρ and L .

By a simple calculation, the condition $\rho \leq \frac{1}{2L}$ leads to

$$\frac{1}{\sqrt{a^2 + b^2}} < -\frac{2b}{a^2 + b^2} \implies a^2 < 3.$$



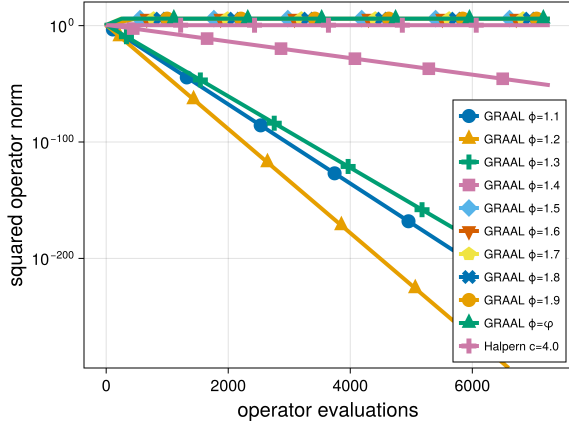
(a) Squared operator norm vs. operator evaluations



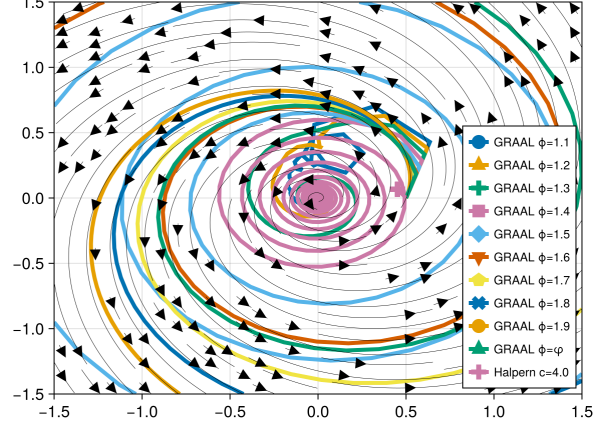
(b) Iterate trajectories over the vector field

Figure 1: Lower bound, Case I: $\rho > \frac{1}{L}$.

As expected, for ϕ ranging from 1.1 to 2, most runs diverged. Interestingly, when $\phi = 1.1$ the method converged with surprisingly good performance. Halpern, on the other hand, diverged in this regime.



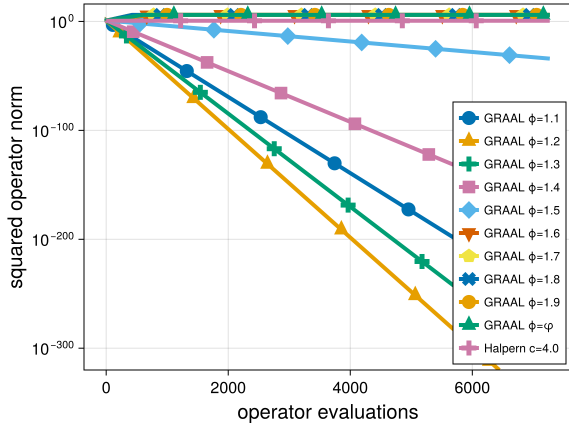
(a) Squared operator norm vs. operator evaluations



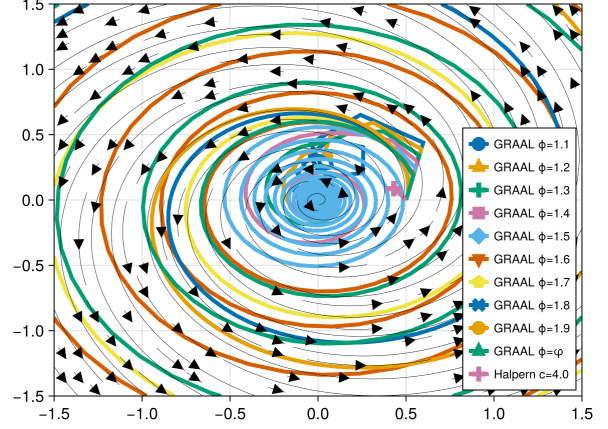
(b) Iterate trajectories over the vector field

Figure 2: Lower bound, Case II: $\frac{1}{2L} < \rho \leq \frac{1}{L}$.

In the intermediate Case II, Halpern converged slowly but steadily. GRAAL also converged when ϕ was relatively small (1.1–1.4), though for larger ϕ values instability reappeared.



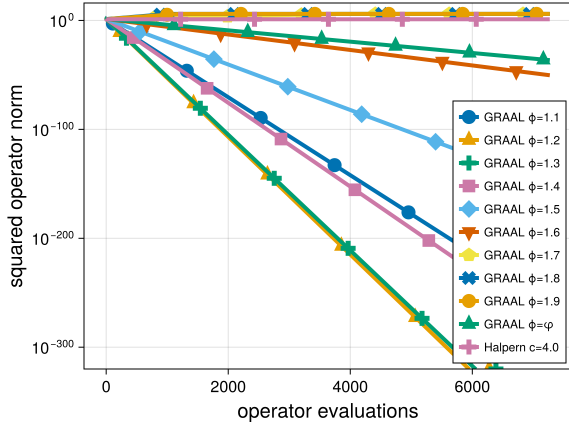
(a) Squared operator norm vs. operator evaluations



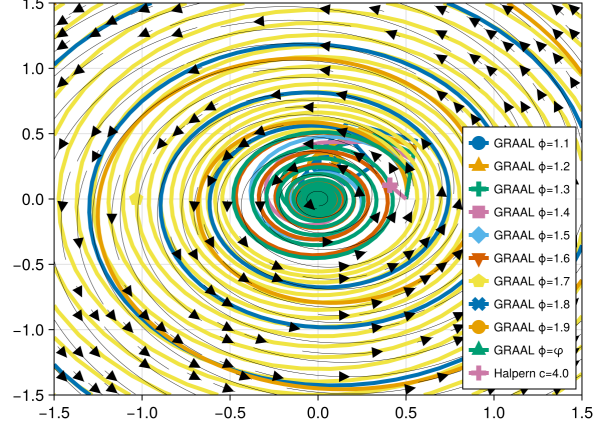
(b) Iterate trajectories over the vector field

Figure 3: Lower bound, Case III: $\rho \leq \frac{1}{2L}$.

In Case III, Halpern consistently converged. For GRAAL, runs with large ϕ such like $\phi = \varphi$ or $\phi = 1.9$ diverged, but for smaller ϕ values convergence was observed.



(a) Squared operator norm vs. operator evaluations



(b) Iterate trajectories over the vector field

Figure 4: Lower bound, Additional experiment: $\rho < \frac{1}{4L}$.

As an additional check, when $\rho < \frac{1}{4L}$ we found that GRAAL with ϕ values close to 2 failed to converge, even though smaller ϕ choices consistently remained stable.

Overall, not surprisingly, Halpern satisfies the theoretical convergence condition for this VI problem, although with slower convergence speed compared to GRAAL. However, GRAAL's performance is highly dependent on the choice of ϕ : even when the theoretical condition is not met, small values of ϕ close to 1 still yield good convergence, while values of ϕ near 2 sometimes cause divergence, even when the condition is satisfied.

2.2 Forsaken

We begin with the standard Forsaken problem

$$\min_x \max_y x(y - a) + f(x) - f(y),$$

where $f(z) = \frac{1}{4}z^2 - \frac{1}{2}z^4 + \frac{1}{6}z^6$. With boundary $[\frac{b}{2}, -\frac{b}{2}]$.

Since both phi and L is sensitive with boundary and parameter, we will tune both case to find those three situation.

We set the boundary to $[-\frac{3}{2}, \frac{3}{2}]$ with $a = 0.45$. This yields

$$L \approx 12.50, \quad \frac{1}{L} \approx 0.08, \quad \rho \approx 0.478, \quad \text{solution} \approx (0.0780, 0.4119).$$

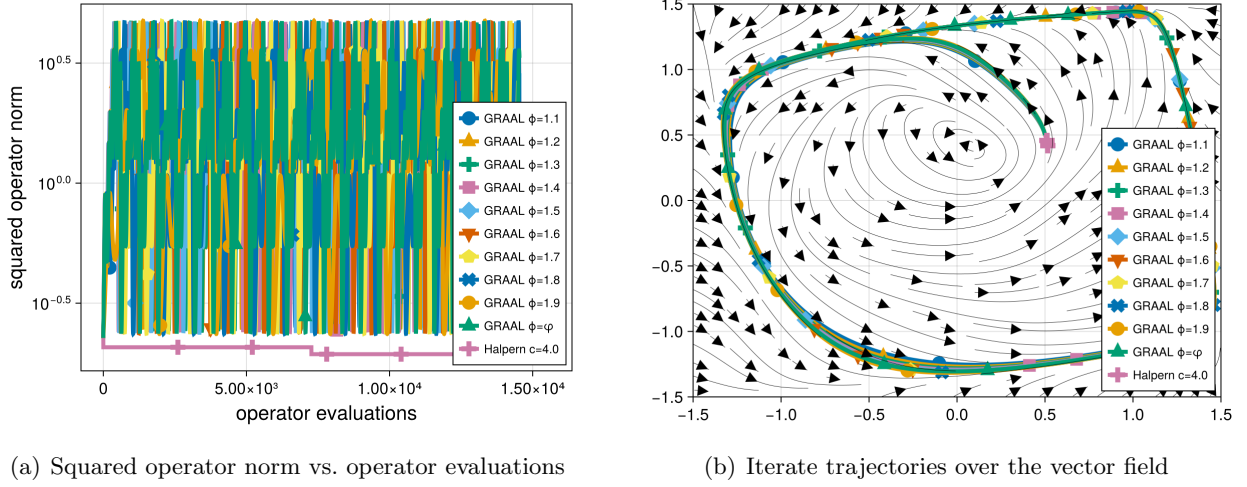
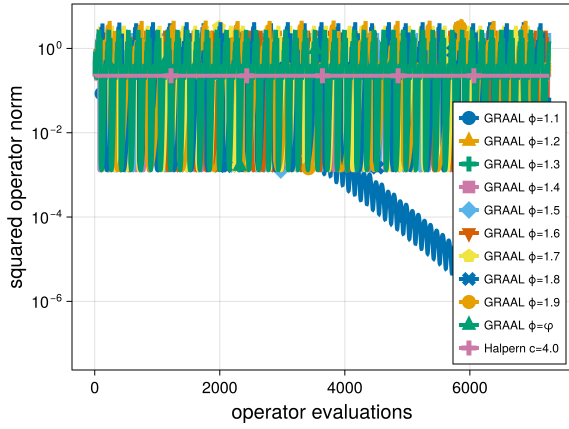


Figure 5: Forsaken, Case I ($\rho > \frac{1}{L}$).

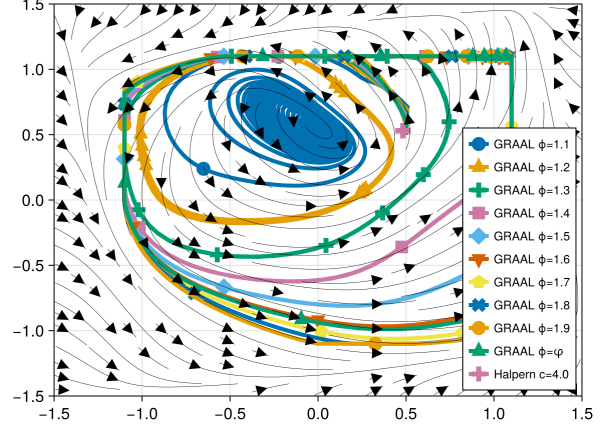
As shown in Fig. 5, Halpern initially showed decay but eventually in later iteration trapped in a limit cycle, while GRAAL consistently diverged for all tested ϕ values.

With boundary $2.2/2$ and $a = 0.55$, we obtain

$$L \approx 1.977, \quad \frac{1}{L} \approx 0.506, \quad \rho \approx 0.409, \quad \text{solution} \approx (-0.0131, 0.5565).$$



(a) Squared operator norm vs. operator evaluations



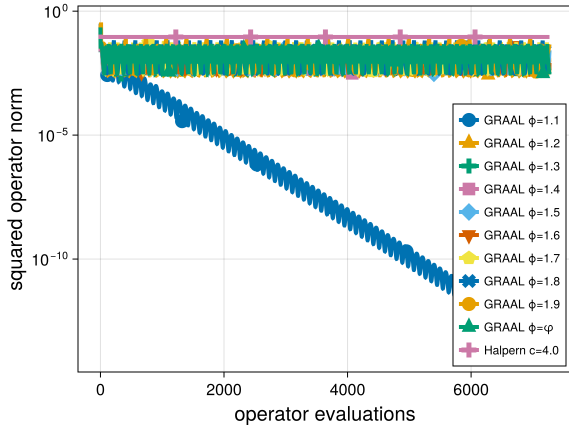
(b) Iterate trajectories over the vector field

Figure 6: Forsaken, Case II ($\frac{1}{2L} < \rho < \frac{1}{L}$).

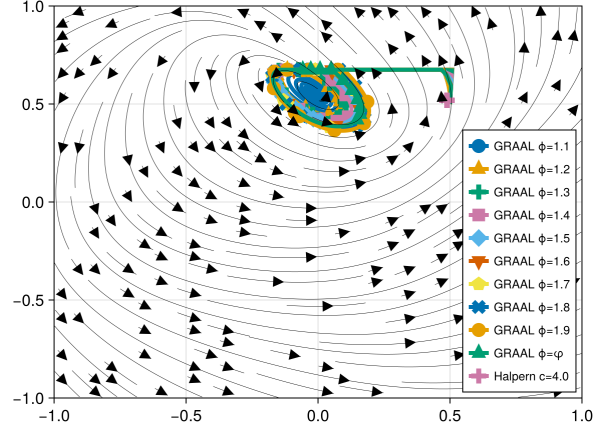
Here Halpern slowly diverged, entering a limit cycle. For GRAAL, only $\phi = 1.1$ led to convergence, while larger ϕ values produced limit cycles without decay.

With boundary $1.35/2$ and $a = 0.55$, we obtain

$$L \approx 1.907, \quad \frac{1}{L} \approx 0.524, \quad \frac{1}{2L} \approx 0.262, \quad \rho \approx 0.249, \quad \text{solution} \approx (0.675, 0.3401).$$



(a) Squared operator norm vs. operator evaluations



(b) Iterate trajectories over the vector field

Figure 7: Forsaken, Case III ($\rho \leq \frac{1}{2L}$).

In this case, Halpern converged reliably, while GRAAL converged only for $\phi = 1.1$. For larger ϕ , the iterates were trapped in limit cycles.

Overall, GRAAL often produced good outcomes even outside its nominal regime: although strict convergence was not guaranteed, the method sometimes escaped limit cycles and reached stable behavior. By contrast, Halpern typically exhibited initial decay, but beyond a certain point the iterates became trapped in limit cycles and failed to achieve convergence.

3 Conclusion

In summary, our experiments highlight that GRAAL can remain effective even when the theoretical condition $\rho \leq \frac{1}{2L}$ is violated. In particular, with ϕ set close to 1.1, GRAAL often converged reliably, escaping limit cycles and stabilizing despite $\rho > \frac{1}{2L}$. By contrast, Halpern showed only transient decay and eventually became trapped in limit cycles outside the strict regime. The key takeaway is that practical performance of GRAAL with small ϕ is more robust than predicted by the theory.